

PQC-Readiness Audit Report

Contract: /mnt/d/Crypto/audit_case_studies/contracts/UniswapV2ERC20.sol
SHA-256: b8ce88929ce2c93b8fce385f8d57701b... **Lines:** 94 **Date:** 2026-08-20 **Tool:** pqc-audit v0.2.0

Executive Summary

Check	Result	Detail
Security Scan	WARN	0 critical, 1 high
PQC Readiness	FAIL	1 quantum-vulnerable primitives, 0 critical implementation advisories
Gas Optimization	3 opportunities	~2500 gas units
OVERALL	FAIL	

Phase 1: Structure

- **Solidity Version:** =0.5.16
- **Contracts:** 1
- **Functions:** 8
- **Events:** 2

Phase 2: Gas Optimization (3 found)

#	Pattern	Impact	Fixable
1	Use custom errors instead of require with string message	Info	Manual
2	Use uint256 instead of smaller uints	Info	Manual
3	Use uint256 instead of smaller uints	Info	Manual

Phase 3: Security Analysis (3 findings)

Severity	Count
Critical	0
High	1
Medium	1
Low	1

Security Score: 84.0/100

Top Findings

- **[MEDIUM]** Line 91 — Missing zero address check
 - Code: `require(recoveredAddress != address(0) && recoveredAddress == owner, 'UniswapV2: INVALID_SIGNATURE');`
 - Missing zero address check for critical operations.
- **[LOW]** Line None — Lack of event emission
 - State-changing functions detected without event emissions
- **[HIGH]** Line None — Improper access control
 - Critical functions found without proper access control

Phase 4: Post-Quantum Cryptography Readiness

Line	Primitive	Risk	Migration Path
90	<code>ecrecover</code>	CRITICAL	Replace with ML-DSA-65 (Dilithium-3) via a FIPS 204 implemen. . .
29	<code>keccak256</code>	MEDIUM	Double hash output length or use SHA-3-512. . .

Quantum Risk Level: CRITICAL

NIST Status: FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) finalized. FIPS 206 (FN-DSA/Falcon) selected for standardization; publication pending.

Implementation Vulnerability Scan (2024-2026 Research)

Vulnerability	Risk	Source
<i>None detected</i>	LOW	No known implementation vulnerabilities match this contract

Evidence Attestation

- **Source SHA-256:** b8ce88929ce2c93b8fce385f8d57701b1a64c600a95317ea5f3561d36297be32
- **Report SHA-256:** 0e76a67f1525c36f7cf962850953584d...

This report was generated mechanically. No LLM inference was used in the security or PQC assessments. PQC-Readiness Audit — pqc-audit v0.2.0